Algebraic Geometry – Exercise Sheet 1

Week 1: Affine algebraic sets, projective space and localisation

Monday and Thursday afternoon sessions; Lectures 1–5

Difficulty. ⋆ routine; ⋆⋆ synthesis; ⋆ ⋆ ⋆ challenge or extension.
Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.

Monday afternoon — material available: Lecture 1 only

Exercise 1. Three points in the affine plane ⋆ CORE

Let

$$S = \{(0,0),(1,0),(0,1)\} \subset \Bbbk^2.$$

(i) Show that

$$\mathbf{I}(S) = (x,y) \cap (x-1,y) \cap (x,y-1).$$

(ii) Prove directly that

$$\mathbf{I}(S) = \big(xy,\, x(x-1),\, y(y-1)\big).$$

(iii) Show that evaluation at the three points induces an isomorphism

$$\Bbbk[x,y]/\mathbf{I}(S) \simeq \Bbbk^3.$$

Exercise 2. Two axes and their coordinate rings ⋆ CORE

In $\Bbbk^2$, set $X = \mathbf{V}(x)$ and $Y = \mathbf{V}(y)$.

(i) Show that

$$X \cup Y = \mathbf{V}(xy), \qquad X \cap Y = \mathbf{V}(x,y).$$

(ii) Compute the coordinate rings of X, Y, $X \cup Y$, and $X \cap Y$.

(iii) Explain geometrically why $\Bbbk[x,y]/(xy)$ has zero divisors.

Exercise 3. The parabola and the cusp ⋆⋆ CORE

Consider

$$P = \mathbf{V}(y - x^2), \qquad C = \mathbf{V}(y^2 - x^3) \subset \Bbbk^2.$$

(i) Show that $t \mapsto (t, t^2)$ defines an isomorphism $\mathbb{A}^1 \to P$.

(ii) Show that $t \mapsto (t^2, t^3)$ defines a morphism $\nu : \mathbb{A}^1 \to C$ which is bijective on the underlying sets of $\Bbbk$-points.

(iii) Write the corresponding homomorphism of coordinate rings and prove that ν is not an isomorphism.

Exercise 4. Polynomial maps of the affine line ⋆ CHOICE

Show that morphisms of affine algebraic sets $\mathbb{A}^1_{\Bbbk} \to \mathbb{A}^1_{\Bbbk}$ are exactly polynomial maps $t \mapsto f(t)$. Describe the coordinate-ring homomorphism and check explicitly how composition reverses arrows.

Thursday afternoon — material available: Lectures 1–4

Exercise 5. Radicals and reduced rings ⋆ CORE

Compute

$$\sqrt{(x^2, y^3)}, \qquad \sqrt{(x^2y^3)}$$

in $\Bbbk[x,y]$. Decide which of the rings

$$\Bbbk[x]/(x^2), \qquad \Bbbk[x,y]/(xy), \qquad \Bbbk[x,y]/(x^2, xy)$$

are reduced.

Exercise 6. Points at infinity ⋆ CORE

In the affine chart $x_0 \neq 0$ of $\mathbb{P}^2_{\Bbbk}$, write $x = x_1/x_0$ and $y = x_2/x_0$.

(i) Homogenise the parabola $y = x^2$ and determine its points at infinity.

(ii) Homogenise the hyperbola $xy = 1$ and determine its points at infinity.

(iii) Explain geometrically why the answers are different.

Exercise 7. The line through two projective points $\star$ CHOICE

Let $p = [a_0 : \cdots : a_n]$ and $q = [b_0 : \cdots : b_n]$ be distinct points of $\mathbb{P}^n_{\Bbbk}$. Show that the line through them is defined by the 3×3 minors of

$$\begin{pmatrix} a_0 & \cdots & a_n \\ b_0 & \cdots & b_n \\ x_0 & \cdots & x_n \end{pmatrix}.$$

Write the equation explicitly for $n = 2$.

Exercise 8. Points in $\mathbb{P}^2$ from minors $\star\star$ CHOICE

Show that the 2×2 minors of

$$M = \begin{pmatrix} x & 0 \\ y & y \\ 0 & z \end{pmatrix}$$

define the three coordinate points in $\mathbb{P}^2$. For four lines $\ell_1, \ldots, \ell_4$ with no three concurrent, show that the maximal minors of

$$\begin{pmatrix} \ell_1 & 0 & 0 \\ -\ell_2 & \ell_2 & 0 \\ 0 & -\ell_3 & \ell_3 \\ 0 & 0 & -\ell_4 \end{pmatrix}$$

define their six pairwise intersection points.

Exercise 9. Equations of the Segre and Veronese images $\star\star\star$ OPTIONAL

a) For

$$([s_0 : s_1], [t_0 : t_1]) \longmapsto [s_0t_0 : s_0t_1 : s_1t_0 : s_1t_1],$$

compute the kernel of $z_{ij} \mapsto s_i t_j$ and identify the image in $\mathbb{P}^3$.

b) For the quadratic Veronese map $\mathbb{P}^2 \to \mathbb{P}^5$, arrange the target coordinates as a symmetric 3×3 matrix and show that the image is defined by its 2×2 minors.

Exercise 10. Localising the coordinate axes $\star$ CORE

Let

$$A = \Bbbk[x, y]/(xy).$$

(i) Compute A_x and A_y.

(ii) Compute $A_{(x)}$ and $A_{(y)}$.

(iii) Explain what these localisations retain and what they discard geometrically.

Exercise 11. Localisation with a nilpotent $\star$ CHOICE

Let $A = \Bbbk[\varepsilon]/(\varepsilon^2)$. Compute

$$A_\varepsilon, \qquad A_{(\varepsilon)}, \qquad A_{1+\varepsilon}.$$

Explain why the three answers are so different.

Exercise 12. Two finite module structures $\star\star$ CHOICE

Let

$$B = \Bbbk[x, y]/(y^2 - x).$$

Show that B is free of rank 2 over $\Bbbk[x]$ and free of rank 1 over $\Bbbk[y]$. Compare geometrically the corresponding morphisms to the affine line.

Exercise 13. Noether normalisation for the hyperbola $\star\star\star$ CHOICE

Let

$$B = \Bbbk[x, y]/(xy - 1), \qquad u = x + y \in B.$$

Show that B is finite over $\Bbbk[u]$. Deduce that the corresponding morphism to $\mathbb{A}^1_{\Bbbk}$ has fibres with at most two points.

After Friday's lecture or homework

Exercise 14. A presheaf which is not a sheaf $\star$ OPTIONAL

Let $X = U \amalg V$ be the disjoint union of two non-empty open subsets. Define a presheaf $\mathcal{F}$ by

$$\mathcal{F}(W) = \mathbb{Z} \quad \text{for } W \neq \varnothing, \qquad \mathcal{F}(\varnothing) = 0,$$

with the evident restriction maps. Show that $\mathcal{F}$ is not a sheaf.

Exercise 15. Continuous and holomorphic functions $\star\star$ CHOICE

Let $U \subset \mathbb{C}$ be connected and contain at least two points. Show that $\mathcal{O}(U)$ has no zero divisors, whereas $\mathcal{C}^0(U)$ has non-zero functions f, g with $fg = 0$.

Exercise 16. $\mathcal{O}(d)$ on the projective line $\star\star\star$ OPTIONAL

Identify $\mathbb{P}^1_{\mathbb{C}}$ with $\mathbb{C} \cup \{\infty\}$. Define $\mathcal{O}(d)$ as the sheaf of meromorphic functions which are holomorphic away from ∞ and have at most a pole of order d there when $d \geq 0$; for $d < 0$, require a zero of order at least $-d$ at ∞. Compute its global sections.

Algebraic Geometry – Exercise Sheet 2

Week 2: Prime spectra, the structure sheaf and points of affine space

Monday and Thursday afternoon sessions; Lectures 6–10

Difficulty. $\star$ routine; $\star\star$ synthesis; $\star\star\star$ challenge or extension.
Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.

Monday afternoon — material available: Lecture 6 only

Exercise 1. The affine line as a spectrum $\star$ **CORE**

Assume that $\Bbbk$ is algebraically closed and let $X = \operatorname{Spec}(\Bbbk[t])$.

(i) List the points of X.

(ii) Determine which points are closed and compute the closure of each point.

(iii) For a non-zero polynomial f, describe the closed subset $\mathbb{V}(f)$.

Exercise 2. The spectrum of the integers $\star$ **CORE**

Describe $\operatorname{Spec}(\mathbb{Z})$: its points, its closed points, and the closure of each point. For $n \neq 0$, determine $\mathbb{V}(n)$.

Exercise 3. Nilpotents are invisible topologically $\star$ **CORE**

Let

$$A = \Bbbk[\varepsilon]/(\varepsilon^2), \qquad X = \operatorname{Spec}(A).$$

Determine the points and open subsets of X. Compare the underlying topological space with $\operatorname{Spec}(\Bbbk)$, and explain why the two affine schemes are nevertheless different.

Exercise 4. A quotient map on spectra $\star$ **CHOICE**

Consider the quotient homomorphism

$$\Bbbk[t] \longrightarrow \Bbbk[t]/(t^2).$$

Describe the induced map on spectra and its image. Why does the image depend only on $\sqrt{(t^2)}$?

Exercise 5. Closed subsets in the node $\star\star$ **CHOICE**

Let $A = \Bbbk[x, y]/(xy)$ and $X = \operatorname{Spec}(A)$. Denote the classes of x, y by the same letters.

(i) Show that $\mathbb{V}(x) \cup \mathbb{V}(y) = X$ and determine $\mathbb{V}(x) \cap \mathbb{V}(y)$.

(ii) Compute the closures of the points (x) and (y).

(iii) Show that neither of these two points is closed.

Thursday afternoon — material available: Lectures 6–9

Exercise 6. The affine line over several fields $\star\star$ **CORE**

Let K be a field and let $X = \operatorname{Spec}(K[u])$.

(i) Describe all points of X, their closures, and their residue fields in terms of irreducible polynomials in $K[u]$.

(ii) Specialise the description to $K = \mathbb{C}$, $K = \mathbb{R}$, and $K = \mathbf{F}_p$. Which closed points are K-rational?

(iii) Take $K = \mathbb{C}(t)$. Give a K-rational closed point and a non-rational closed point, and compute their residue fields.

Exercise 7. Principal opens in the node $\star\star$ **CORE**

Let $A = \Bbbk[x, y]/(xy)$ and $X = \operatorname{Spec}(A)$.

(i) Describe $D(x)$ and $D(y)$ as affine schemes.

(ii) Show that $D(x) \cap D(y) = \varnothing$ and identify the point missing from $D(x) \cup D(y)$.

(iii) Compute the residue fields at (x), (y) and (x, y).

Exercise 8. Sections and a stalk on the affine line ⋆ CORE

Let $X = \operatorname{Spec}(\Bbbk[t])$ and $U = D(t(t-1))$.

(i) Compute $\mathcal{O}_X(U)$.

(ii) Compute the stalk $\mathcal{O}_{X,(t-a)}$ and its residue field.

(iii) For which a does $(t-a)$ belong to U?

Exercise 9. A subscheme of the affine line with two support points ⋆⋆ CORE

Let

$$Z = \operatorname{Spec}\big(\Bbbk[t]/(t^3(t-1)^2)\big).$$

(i) Determine the underlying topological space of Z.

(ii) Decompose its coordinate ring as a product of two rings.

(iii) Compare the two local pieces with the reduced points at 0 and 1.

Exercise 10. Prime ideals and scheme structure in the affine plane ⋆⋆ CORE

Assume that $\Bbbk$ is algebraically closed and work in $A = \Bbbk[x,y]$. Put

$$I = (xy), \qquad J = (x^2y), \qquad Q = (x,y)^2.$$

(i) Using the description of the prime ideals of A, determine the primes containing I, J, and Q. Hence compare the three underlying closed subsets.

(ii) Verify in these cases the formula

$$\sqrt{\mathfrak{a}} = \bigcap_{\mathfrak{p}\supseteq\mathfrak{a}} \mathfrak{p}.$$

(iii) Explain how $\operatorname{Spec}(A/I)$, $\operatorname{Spec}(A/J)$, and $\operatorname{Spec}(A/Q)$ retain different scheme structures even when their points coincide.

Exercise 11. Products of rings and disjoint unions ⋆⋆ CHOICE

Let B and C be non-zero rings.

(i) Show that every prime ideal of $B \times C$ is of the form $\mathfrak{p} \times C$ or $B \times \mathfrak{q}$.

(ii) Deduce that

$$\operatorname{Spec}(B \times C) \simeq \operatorname{Spec}(B) \amalg \operatorname{Spec}(C).$$

(iii) Relate this to the idempotents $(1,0)$ and $(0,1)$.

After Friday's arithmetic lecture or homework

Exercise 12. Prime ideals in $\mathbb{Z}[\sqrt{3}]$ ⋆⋆ CHOICE

Let

$$A = \mathbb{Z}[s]/(s^2-3), \qquad f : \operatorname{Spec}(A) \longrightarrow \operatorname{Spec}(\mathbb{Z}).$$

For a prime p, describe the points of $\operatorname{Spec}(A)$ mapping to (p) by factoring s^2-3 over $\mathbf{F}_p$. Treat separately $p=2$, $p=3$, and $p \neq 2,3$.

Exercise 13. An arithmetic conic ⋆⋆⋆ OPTIONAL

For

$$X = \operatorname{Spec}\big(\mathbb{Z}[x,y]/(x^2+y^2-5)\big) \longrightarrow \operatorname{Spec}(\mathbb{Z}),$$

write down the equation obtained after reduction modulo a prime p. Determine for which p that equation is a square or has a singular point.

Algebraic Geometry – Exercise Sheet 3

Week 3: Schemes, morphisms, gluing and first global properties

Monday and Thursday afternoon sessions; Lectures 11–15

Difficulty. $\star$ routine; $\star\star$ synthesis; $\star\star\star$ challenge or extension.
Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.

Monday afternoon — material available: Lectures 1–11

Exercise 1. Global functions as morphisms to the affine line $\star$ **CORE**

Let X be a scheme.

(i) Choose an affine open covering $X = \bigcup_i U_i$. Explain why a global section $s \in \Gamma(X, \mathcal{O}_X)$ determines compatible morphisms $U_i \to \mathbb{A}^1_{\mathbb{Z}}$ and hence a morphism

$$f_s : X \longrightarrow \mathbb{A}^1_{\mathbb{Z}}.$$

(ii) Show that

$$\mathrm{Hom}(X, \mathbb{A}^1_{\mathbb{Z}}) \simeq \Gamma(X, \mathcal{O}_X).$$

(iii) If $X = \mathrm{Spec}(A)$, describe the morphism associated with $a \in A$.

Exercise 2. The reduced point and the double point $\star$ **CORE**

Inside $\mathbb{A}^1_{\Bbbk} = \mathrm{Spec}(\Bbbk[t])$, consider

$$P = \mathrm{Spec}(\Bbbk[t]/(t)), \qquad D = \mathrm{Spec}(\Bbbk[t]/(t^2)).$$

(i) Show that the two closed immersions have the same image as subsets of $\mathbb{A}^1_{\Bbbk}$.

(ii) Show that P and D are not isomorphic as schemes.

(iii) Compute the maps on local rings at their unique points.

Exercise 3. A morphism between affine schemes $\star$ **CORE**

Let

$$\varphi : \Bbbk[u, v] \longrightarrow \Bbbk[t], \qquad u \longmapsto t, \quad v \longmapsto t^2.$$

(i) Describe the induced morphism $f : \mathbb{A}^1_{\Bbbk} \to \mathbb{A}^2_{\Bbbk}$.

(ii) Determine $\ker(\varphi)$ and factor f through a closed subscheme of $\mathbb{A}^2_{\Bbbk}$.

(iii) Show that the resulting morphism onto that closed subscheme is an isomorphism.

Exercise 4. Gluing ideal sheaves and closed subschemes $\star\star$ **CORE**

Let $X = \bigcup_i U_i$ be an open covering. Suppose that each U_i carries a quasi-coherent ideal sheaf $\mathcal{I}_i \subset \mathcal{O}_{U_i}$ and that

$$\mathcal{I}_i|_{U_i \cap U_j} = \mathcal{I}_j|_{U_i \cap U_j}$$

inside $\mathcal{O}_X|_{U_i \cap U_j}$.

(i) Show that the $\mathcal{I}_i$ glue to an ideal sheaf $\mathcal{I} \subset \mathcal{O}_X$.

(ii) Prove that $\mathcal{I}$ is quasi-coherent.

(iii) If $Z_i \hookrightarrow U_i$ is the closed subscheme defined by $\mathcal{I}_i$, show that the Z_i glue to the closed subscheme of X defined by $\mathcal{I}$.

(iv) Explain why none of these assertions requires X to be affine.

Exercise 5. Open subschemes and the punctured plane $\star\star$ **CORE**

Let X be a scheme and let $U \subset X$ be open.

(i) Prove directly from the locally affine definition that $(U, \mathcal{O}_X|_U)$ is a scheme.

(ii) Show that for every $x \in U$ one has

$$\mathcal{O}_{U,x} \simeq \mathcal{O}_{X,x}, \qquad \kappa_U(x) \simeq \kappa_X(x).$$

(iii) Cover the punctured affine plane $P = \mathbb{A}^2_{\Bbbk} \setminus \mathbb{V}(x, y)$ by two affine open subschemes.

(iv) Compute $\Gamma(P, \mathcal{O}_P)$ and prove that P is not affine.

Thursday afternoon — material available: Lectures 1–14

Exercise 6. Dual numbers and tangent vectors ⋆⋆ CORE

Let $D = \operatorname{Spec}(\Bbbk[\varepsilon]/(\varepsilon^2))$ and let $p = (a, b) \in \mathbb{A}^2_{\Bbbk}(\Bbbk)$.

(i) Show that morphisms $D \to \mathbb{A}^2_{\Bbbk}$ whose closed point maps to p are uniquely of the form

$$x \longmapsto a + u\varepsilon, \qquad y \longmapsto b + v\varepsilon.$$

(ii) Let $C = \mathbb{V}(f) \subset \mathbb{A}^2_{\Bbbk}$. Show that such a morphism factors through C precisely when

$$f(a, b) = 0, \qquad f_x(a, b)u + f_y(a, b)v = 0.$$

(iii) Apply this to the parabola $y = x^2$.

Exercise 7. The line with two origins ⋆⋆ CORE

Glue two copies

$$X_1 = \operatorname{Spec}(\Bbbk[t_1]), \qquad X_2 = \operatorname{Spec}(\Bbbk[t_2])$$

along $D(t_1) \simeq D(t_2)$ by $t_1 \mapsto t_2$. Let X be the resulting scheme.

(i) Describe the underlying topological space.

(ii) Show that $\Gamma(X, \mathcal{O}_X) \simeq \Bbbk[t]$.

(iii) Show that the canonical map $X \to \operatorname{Spec} \Gamma(X, \mathcal{O}_X)$ identifies the two origins.

(iv) Deduce that X is not affine.

Exercise 8. Rational and closed points of the projective line ⋆⋆ CORE

Construct $\mathbb{P}^1_{\Bbbk}$ by gluing $U_0 = \operatorname{Spec}(\Bbbk[t])$ and $U_1 = \operatorname{Spec}(\Bbbk[s])$ along $s = t^{-1}$.

(i) For every field extension $K/\Bbbk$, prove that

$$\mathbb{P}^1_{\Bbbk}(K) = \mathbf{P}^1_K.$$

Describe the points in the two standard charts.

(ii) Describe all closed points of $\mathbb{P}^1_{\Bbbk}$, compute their residue fields, and determine which are $\Bbbk$-rational.

(iii) Prove that $\Gamma(\mathbb{P}^1_{\Bbbk}, \mathcal{O}_{\mathbb{P}^1}) = \Bbbk$.

(iv) Deduce that $\mathbb{P}^1_{\Bbbk}$ is not affine.

Exercise 9. The projective conic from its standard opens ⋆⋆ CORE

Let

$$S = \Bbbk[x_0, x_1, x_2]/(x_0x_2 - x_1^2), \qquad C = \operatorname{Proj}(S).$$

(i) Compute $(S_{x_0})_0$ and $(S_{x_2})_0$.

(ii) Identify $D_+(x_0)$ and $D_+(x_2)$ with affine lines.

(iii) Describe their overlap and recover the gluing rule for $\mathbb{P}^1$.

Exercise 10. Connectedness beyond affine schemes ⋆⋆ CORE

Let X be a scheme.

(i) Prove that X is disconnected if and only if $\Gamma(X, \mathcal{O}_X)$ has a non-trivial idempotent.

(ii) Use the global-function computations from this sheet to show that $\mathbb{P}^1_{\Bbbk}$ and the line with two origins are connected.

(iii) Compare the connected reducible scheme $\mathrm{Spec}(\Bbbk[x,y]/(xy))$ with the disconnected scheme $\mathrm{Spec}(\Bbbk \times \Bbbk)$.

Exercise 11. A doubled projective line and its reduction $\star\star\star$ **CORE**

Let

$$U_0 = \mathrm{Spec}\big(\Bbbk[t,\varepsilon]/(\varepsilon^2)\big), \qquad U_1 = \mathrm{Spec}\big(\Bbbk[s,\eta]/(\eta^2)\big).$$

Glue $D(t) \subset U_0$ to $D(s) \subset U_1$ using $s = t^{-1}$ and $\eta = \varepsilon$, and call the resulting scheme X.

(i) Show that $(\varepsilon) \subset \mathcal{O}_{U_0}$ and $(\eta) \subset \mathcal{O}_{U_1}$ glue to a quasi-coherent ideal sheaf $\mathcal{I} \subset \mathcal{O}_X$ with $\mathcal{I}^2 = 0$.

(ii) Describe X_{red} and prove that $X_{\mathrm{red}} \simeq \mathbb{P}^1_{\Bbbk}$.

(iii) Show that every morphism from a reduced scheme Y to X factors uniquely through $X_{\mathrm{red}} \hookrightarrow X$.

(iv) Deduce that X is not affine.

Exercise 12. A weighted projective chart $\star\star\star$ **OPTIONAL**

Let $R = \Bbbk[x,y,z]$ with $\deg x = \deg y = 1$ and $\deg z = 2$, and set $X = \mathrm{Proj}(R)$. Show that $D_+(x)$ and $D_+(y)$ are affine planes and that

$$D_+(z) \simeq \mathrm{Spec}\big(\Bbbk[a,b,c]/(ac - b^2)\big).$$

Identify the singular point of this chart.

After Friday's lecture or preparation for Week 4

Exercise 13. Components and generic points $\star\star$ **CHOICE**

Let

$$I = (xy, xz) \subset \Bbbk[x,y,z], \qquad X = \mathrm{Spec}(\Bbbk[x,y,z]/I).$$

Show that $I = (x) \cap (y,z)$. Determine the irreducible components, their generic points and dimensions, and decide whether X is connected or integral.

Exercise 14. The function field of the cusp $\star\star$ **CHOICE**

Let

$$C = \mathrm{Spec}\big(\Bbbk[x,y]/(y^2 - x^3)\big).$$

Show that C is integral and that $K(C) \simeq \Bbbk(t)$.

Algebraic Geometry – Exercise Sheet 4

Week 4: Generic points, dimension and scheme-theoretic fibres

One Monday exercise session, followed by a revision bank; Lectures 15–18

Difficulty. $\star$ routine; $\star\star$ synthesis; $\star\star\star$ challenge or extension.
Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.

Monday afternoon — material available: Lectures 1–16

Exercise 1. Components and generic points $\star\star$ **CORE**

Let

$$I = (xy, xz) \subset \Bbbk[x, y, z], \qquad X = \mathrm{Spec}(\Bbbk[x, y, z]/I).$$

(i) Show that $I = (x) \cap (y, z)$.

(ii) Determine the irreducible components and their generic points.

(iii) Is X reduced, connected, irreducible, or integral?

Exercise 2. The function field of the cusp $\star\star$ **CORE**

Let

$$C = \mathrm{Spec}\big(\Bbbk[x, y]/(y^2 - x^3)\big).$$

Show that C is integral, compute its function field, and deduce its dimension.

Exercise 3. Chains and dimensions $\star$ **CORE**

(i) Exhibit a maximal chain of irreducible closed subsets in $\mathrm{Spec}(\mathbb{Z})$.

(ii) Give a chain of length n in $\mathbb{A}^n_{\Bbbk}$.

(iii) Determine the dimension of

$$\mathrm{Spec}\big(\Bbbk[x, y, z]/(xy, xz)\big)$$

from its components.

Exercise 4. Noether normalisation and dimension $\star\star$ **CORE**

Let

$$B = \Bbbk[x, y]/(y^2 - x^3).$$

(i) Show that B is finite over $\Bbbk[x]$.

(ii) Use this to recover $\dim(B) = 1$.

(iii) Compare with the computation from the function field.

After Lectures 17–18 — revision bank on fibres

Exercise 5. The generic affine line $\star$ **CORE**

Let

$$\pi : \mathbb{A}^2_{\mathbb{C}} = \mathrm{Spec}(\mathbb{C}[t, x]) \longrightarrow \mathbb{A}^1_{\mathbb{C}} = \mathrm{Spec}(\mathbb{C}[t])$$

be the projection. Compute its fibre over a closed point $(t-a)$ and its fibre over the generic point (0). Explain why the latter is $\mathrm{Spec}(\mathbb{C}(t)[x])$ and verify the dimension formula for the generic fibre.

Exercise 6. A fibre product larger than the set-theoretic one $\star\star$ **CHOICE**

Let $X = \mathrm{Spec}(\mathbb{C})$ and $S = \mathrm{Spec}(\mathbb{R})$. Compute $X \times_S X$ and compare its points with the set-theoretic fibre product of the underlying one-point spaces.

Exercise 7. The power map and its fibres $\star\star$ **CORE**

Assume that $\Bbbk$ is algebraically closed of characteristic 0. Let

$$f : \mathbb{A}^1_{\Bbbk} \longrightarrow \mathbb{A}^1_{\Bbbk}, \qquad t \longmapsto t^n.$$

Compute every scheme-theoretic fibre, determine when it is reduced, compute the generic fibre, and show that f is finite.

Exercise 8. A family whose special fibre becomes reducible ⋆⋆ CORE

Let

$$X = \operatorname{Spec}\bigl(\Bbbk[x,y,t]/(xy-t)\bigr) \longrightarrow \mathbb{A}^1_t.$$

Compute the fibres, show that all have dimension 1, and determine which are integral and which are reducible.

Exercise 9. A jumping fibre ⋆⋆ CORE

Consider

$$f : \mathbb{A}^2_{\Bbbk} \longrightarrow \mathbb{A}^2_{\Bbbk}, \qquad (x,y) \longmapsto (x, xy).$$

Compute the fibre over (a,b) and show that the generic fibre has dimension 0, while the fibre over $(0,0)$ has dimension 1.

Exercise 10. The generic fibre formula in examples ⋆⋆ CHOICE

Verify

$$\dim X_\eta = \dim X - \dim Y$$

for the map in the preceding exercise and for the projection $\mathbb{A}^{m+n}_{\Bbbk} \to \mathbb{A}^m_{\Bbbk}$.

Optional outreach exercises: Hilbert schemes of points

Exercise 11. $\mathrm{Hilb}^n(\mathbb{A}^1)$ ⋆⋆ CHOICE

Assume that $\Bbbk$ is algebraically closed. Show that every ideal $I \subset \Bbbk[x]$ with $\dim_{\Bbbk} \Bbbk[x]/I = n$ is generated by a unique monic polynomial of degree n. Deduce

$$\mathrm{Hilb}^n(\mathbb{A}^1_{\Bbbk}) \simeq \mathbb{A}^n_{\Bbbk}$$

and explain how factorisation records support and multiplicities.

Exercise 12. Commuting matrices and a cyclic vector ⋆⋆⋆ OPTIONAL

Let V be n-dimensional and let (X,Y,v) satisfy $[X,Y]=0$ and $\Bbbk[X,Y]v = V$. Show that

$$I = \{f \in \Bbbk[x,y] \mid f(X,Y)v = 0\}$$

is an ideal and $\Bbbk[x,y]/I \simeq V$. Determine the triple for $I = (y, x^2)$.